useless. The conclusion of Riemann's thesis was that the properties of a curved space are captured by a particular mathematical object which we know today as Riemann's curvature, and indicate with the letter 'R'. Einstein wrote an equation which says that R is equivalent to the energy of matter. That is to say: space curves where there is matter. That is it. The equation fits into half a line, and there is nothing more. A vision – that space curves – became an equation.

But within this equation there is a teeming universe. And here the magical richness of the theory opens up into a phantasmagorical succession of predictions that resemble the delirious ravings of a madman, but which have all turned out to be true.

To begin with, the equation describes how space bends around a star. Due to this curvature, not only do planets orbit round the star, but light stops moving in a straight line and deviates. Einstein predicted that the sun causes light to deviate. In 1919 this deviation was measured, and the prediction verified. But it isn't only space that curves; time does too. Einstein predicted that time passes more quickly high up than below, nearer to the Earth. This was measured and turned out to be the case. If a man who has lived at sea level meets up with his twin who has lived in the mountains, he will find that his sibling is slightly older than him. And this is just the beginning.

When a large star has burnt up all of its combustible substance (hydrogen) it goes out. What remains is no longer supported by the heat of the combustion and collapses under its own weight, to a point where it bends space to such a degree that it plummets into an actual hole. These are the famous 'black holes'. When I was studying at university they were considered to be the barely credible predictions of an esoteric theory. Today they are observed in the sky in their hundreds, and studied in great detail by astronomers.

But this is still not all. The whole of space can expand and contract. Furthermore, Einstein's equation shows that space cannot stand still; it *must be* expanding. In 1930 the expansion of the universe was actually observed. The same equation predicts that the expansion ought to have been triggered by the explosion of a young, extremely small and extremely hot universe: by what we now know as the 'Big Bang'. Once again, no one believed this at first, but the proof mounted up until *cosmic background radiation* – the diffuse glare that remains from the heat generated by the original explosion – was actually observed in the sky. The prediction arising from Einstein's